

A Technical Deep Dive into Amazon S3 Vectors Bucket

TABLE OF CONTENTS

[Architecting the Future of AI: A Technical White Paper on Amazon S3 Vectors](#)

[Executive Summary](#)

[1. Introduction: The Unstructured Data Revolution and the Rise of Vector Embeddings](#)

[2. The Problem Statement: Deconstructing the Challenges of Traditional Vector Search Architectures](#)

[2.1 Architectural Complexity and Data Synchronization Overheads](#)

[2.2 Prohibitive Operational Costs](#)

[2.3 Scalability and Management Burdens](#)

[3. The Solution: A Technical Deep Dive into Amazon S3 Vectors](#)

[3.1 Core Architecture and Components](#)

[3.1.1 S3 Vector Buckets](#)

[3.1.2 Vector Indexes](#)

[3.1.3 Native Indexing and Search API](#)

[3.2 Seamless Integration with the AWS Ecosystem](#)

[4. The Value Proposition: Key Benefits and Advantages of S3 Vectors](#)

[4.1 Radical Architectural Simplification](#)

[4.2 Unprecedented Cost Reduction](#)

[4.3 Massive Scalability and Enterprise-Grade Durability](#)

[4.4 Accelerated AI Development and Innovation](#)

[5. Primary Use Cases and Implementation Patterns](#)

[5.1 Retrieval-Augmented Generation \(RAG\)](#)

[5.2 Enterprise Semantic Search](#)

[5.3 AI Agent Long-Term Memory](#)

[5.4 Multimedia Similarity Search and Analysis](#)

[5.5 Cost-Optimized Tiered Data Architectures](#)

[6. Limitations](#)

[6.1 Preview Release and Service Maturity](#)

[6.2 Account, Region, and Availability Limitations](#)

[6.3 Index Capacity and Vector Configuration](#)

[6.4 Metadata and Filtering Restrictions](#)

[6.5 Write Throughput and Update Model Limitations](#)

[6.6 Latency and Query Performance Boundaries](#)

[6.7 API and Data Size Constraints](#)

[6.8 Summary of Limitations](#)

[7. Technical Considerations and Performance Profile](#)

[7.1 Architectural Philosophy](#)

[7.2 Query Latency and Throughput Performance](#)



[7.3 Storage Design and Index Management](#)
[7.4 Data Type and Computation Efficiency](#)
[7.5 Configuration Immutability Considerations](#)
[7.6 Metadata Filtering and Query Semantics](#)
[7.7 Scalability Planning](#)
[7.8 Balancing Cost, Latency, and Usability](#)
[7.9 Regional and Regulatory Considerations](#)
[Summary: Strategic Perspective](#)
[8. Architectural Best Practices and Integration Patterns](#)
[8.1 Architecture Overview](#)
[8.2 Best Practices for Data Preparation and Embedding](#)
[8.3 AWS Lambda as the Orchestration Layer](#)
[8.4 Leveraging Amazon Bedrock for RAG Workloads](#)
[8.5 Training and Custom Refinement with SageMaker](#)
[8.6 Metadata Management and Auxiliary Indexing](#)
[8.7 Sharding and Query Federation Patterns](#)
[8.8 Security and Compliance Practices](#)
[8.9 Monitoring, Logging, and Cost Optimization](#)
[8.10 Example End-to-End Reference Architecture](#)
[8.11 Future-Ready Design Considerations](#)
[8.12 Synthesis: Designing for Scale, Simplicity, and Continuity](#)
[9. Conclusion: Redefining AI-Ready Storage for the Enterprise](#)

Architecting the Future of AI: A Technical White Paper on Amazon S3 Vectors

Executive Summary

In 2025, Amazon Web Services (AWS) launched a transformative enhancement to its flagship cloud object storage service with the introduction of S3 Vectors. This new capability fundamentally redefines the role of Amazon S3, evolving it into a powerful, highly scalable, and exceptionally cost-effective solution for storing and querying vector embeddings. Vector embeddings are the numerical representations of unstructured data that form the backbone of modern artificial intelligence (AI) and machine learning (ML) applications. S3 Vectors directly addresses the escalating industry demand for efficient, large-scale similarity search capabilities, which are essential for applications like semantic search, recommendation engines, and the increasingly prevalent Retrieval-Augmented Generation (RAG) architectural pattern.

The primary challenge S3 Vectors solves is the inherent complexity and high cost associated with traditional vector search architectures. Historically, implementing vector search necessitated deploying and managing separate, specialized vector databases. This approach introduced significant architectural complexity through multi-layered systems, required intricate data synchronization pipelines, and incurred substantial operational costs, often due to the reliance on memory-intensive infrastructure. S3 Vectors obviates this requirement by deeply integrating vector indexing and search functionalities directly into the S3 storage layer. This strategic integration streamlines AI application architecture, drastically reduces operational overhead, and, according to AWS, can lower the total cost of ownership (TCO) for vector storage and querying by up to 90% compared to conventional solutions.

Objective

This white paper provides a comprehensive technical analysis of Amazon S3 Vectors, designed for AI/ML engineers, solutions architects, and technical project managers. It delves into the underlying architecture, core components, and principal functionalities of this innovative service. The document meticulously explores the significant benefits, including architectural simplification, cost reduction, and seamless integration with the broader AWS ecosystem, particularly with services like Amazon Bedrock, Amazon SageMaker, and Amazon OpenSearch Service. Furthermore, it presents a detailed examination of primary use cases and implementation patterns, offering practical guidance for leveraging S3 Vectors in enterprise-grade AI solutions. Finally, the paper provides a transparent assessment of the service's technical considerations and limitations to empower technical professionals to make informed architectural decisions, ensuring they can harness the full potential of S3 Vectors to build the next generation of scalable, intelligent, and cost-efficient applications on the AWS cloud.

1. Introduction: The Unstructured Data Revolution and the Rise of Vector Embeddings

We are in the midst of a data revolution, defined not by the volume of data alone but by its nature. The vast majority of new data is unstructured—a rich tapestry of text, images, audio, video, and other complex formats. This surge has been catalyzed by the proliferation of AI and machine learning, which provide the tools to unlock the immense value latent within this data. At the heart of this transformation lies a powerful concept: vector embeddings.

Vector embeddings are dense, numerical representations of unstructured data generated by machine learning models. These high-dimensional vectors are meticulously crafted to capture the semantic essence, context, and relationships within the source content. In a vector space, words, sentences, or documents with similar meanings are located closer together. This enables a paradigm shift in data interaction, moving beyond keyword matching to sophisticated, context-aware similarity searches—the engine behind intelligent applications such as hyper-personalized recommendations, semantic search, and Retrieval-Augmented Generation (RAG) models that enhance Large Language Models (LLMs) with external knowledge.

However, the power of vector embeddings introduced infrastructural challenges. Storing, indexing, and querying billions or trillions of high-dimensional vectors efficiently at scale required specialized technology: the vector database. These databases, while effective, added complexity to the stack. Organizations found themselves managing a split architecture—object storage like Amazon S3 for raw data and a separate, often costly, vector database for embeddings. This separation necessitated complex and brittle data pipelines for synchronization, increased operational overhead, and steep costs as data volumes and query loads grew.

Recognizing this friction point in modern AI development, Amazon Web Services engineered a more integrated solution. In 2025, AWS introduced Amazon S3 Vectors, a native capability within the world's most popular object storage service. This feature simplifies the entire vector data workflow by allowing developers to store, index, and query embeddings directly alongside their source data in S3. By doing so, AWS democratizes scalable vector search technology, making it more accessible, architecturally simple, and cost-effective for a broad spectrum of AI applications and organizations. This white paper provides a definitive technical examination of this groundbreaking service.

2. The Problem Statement: Deconstructing the Challenges of Traditional Vector Search Architectures

Before the advent of integrated solutions like S3 Vectors, the implementation of vector search capabilities at an enterprise scale was a formidable engineering endeavor, fraught with challenges that spanned architecture, cost, scalability, and operations. Understanding these hurdles is essential to appreciating the strategic value proposition offered by S3 Vectors. The traditional approach, which relied on coupling a primary object store with a dedicated vector database, presented several significant points of friction.

2.1 Architectural Complexity and Data Synchronization Overheads

The most immediate challenge was the inherent architectural complexity of a multi-system setup. A typical stack involved at least two distinct data stores: a service like Amazon S3 holding the canonical source data (e.g., PDF documents, images, audio files), and a separate, specialized vector database (whether open-source or a managed service) to house the corresponding vector embeddings.

This separation forced the creation and maintenance of complex data pipelines responsible for a multi-step workflow:

- **Data Ingestion:** New or updated objects in S3 would trigger an event.
- **Embedding Generation:** An ETL (Extract, Transform, Load) process would fetch the object, pass it through a machine learning model to generate its vector embedding.
- **Vector Ingestion:** The resulting embedding, along with its associated metadata and a reference back to the original S3 object, would then be written to the vector database.
- **Synchronization and Consistency:** A robust mechanism was required to handle updates and deletions, ensuring that the vector database remained a consistent and accurate index of the data in S3.

These pipelines introduced numerous potential points of failure, increased development and testing cycles, and demanded constant maintenance and monitoring to ensure data integrity. The logical coupling of two physically separate systems created a brittle architecture that was difficult to evolve and scale.

2.2 Prohibitive Operational Costs

A second major barrier was the significant operational cost associated with high-performance vector databases. To deliver the low-latency query performance required by many applications, these databases often rely on memory-intensive infrastructure. Storing billions of high-dimensional vectors in RAM is expensive, and costs scale directly with the size of the dataset and the desired query throughput.

This led to several financial challenges:

- **High Compute Costs:** The need for powerful server instances with large amounts of RAM and fast CPUs resulted in a high, fixed baseline cost, regardless of actual query volume.
- **Idle Capacity:** Organizations were often forced to provision infrastructure for peak query loads, leading to significant costs associated with idle capacity during off-peak hours.
- **Storage vs. Compute Scaling:** In many vector database architectures, storage and compute resources are tightly coupled. This means that to store more vectors, one must also scale up compute resources, even if the query rate does not increase, leading to inefficient resource allocation.

According to AWS analysis, a traditional self-managed vector database hosting just 10 million vectors could incur costs of at least \$300 per month for compute resources alone, before factoring in storage, data transfer, and maintenance overhead.

2.3 Scalability and Management Burdens

While designed for performance, dedicated vector databases can encounter their own scalability limits, especially when datasets grow into the billions or trillions of vectors. Scaling these systems often requires deep expertise in distributed systems and the specific database's sharding and replication mechanisms.

Furthermore, the management overhead was a significant drain on engineering resources:

- **Infrastructure Provisioning:** Teams were responsible for provisioning the correct type and number of servers.
- **Configuration and Tuning:** This included configuring the database, setting up networking and security, and meticulously tuning complex indexing parameters to balance recall, latency, and memory usage.

- **Maintenance:** Ongoing tasks such as patching, upgrades, backups, and disaster recovery planning diverted critical engineering time away from core application development and innovation.

These collective challenges—architectural complexity, prohibitive costs, and significant management overhead—created a substantial barrier to entry, particularly for smaller organizations, and acted as a drag on the scalability and cost-efficiency of AI applications for even the largest enterprises. It was this set of well-understood problems that S3 Vectors was engineered to solve.

3. The Solution: A Technical Deep Dive into Amazon S3 Vectors

Amazon S3 Vectors represents a paradigm shift in how vector embeddings are stored and managed in the cloud. It is a new feature that natively integrates high-performance vector indexing and search directly into the Amazon S3 service. This approach allows developers to store, index, and query vector embeddings alongside their original source data within the same storage layer, effectively unifying the AI data lifecycle and streamlining the entire application architecture.

The fundamental innovation is the introduction of a new S3 bucket type specifically designed for this purpose: the **S3 Vector bucket**. This new primitive is built upon a serverless architecture that separates storage and compute, enabling each to scale independently and eliminating the need for users to provision or manage any underlying infrastructure.

3.1 Core Architecture and Components

The S3 Vectors feature is comprised of several key components that work in concert to provide a seamless vector search experience.

3.1.1 S3 Vector Buckets

At the highest level is the Vector Bucket, a specialized type of S3 bucket purpose-built for the efficient storage and management of vector embeddings. While they exist within the S3 ecosystem, Vector Buckets are distinct from standard S3 buckets and are accessed via a dedicated set of APIs. An AWS account is permitted to create up to 10,000 vector buckets per region, providing ample capacity for organizing different projects, applications, or data domains.

3.1.2 Vector Indexes

Within each Vector Bucket, users can create multiple **Vector Indexes** to organize their embeddings. An index is the core structure that contains the vectors and facilitates efficient similarity search. Each bucket can contain up to 10,000 indexes. When creating an index, several critical parameters must be defined, and these parameters are **immutable** once the index is created:

- **Vector Dimension:** The dimensionality of the vectors that will be stored in the index. S3 Vectors supports dimensions from 1 to 4,096. This must match the output dimension of the embedding model being used.
- **Distance Metric:** The mathematical function used to measure the similarity or distance between two vectors. S3 Vectors supports the most common metrics used in AI applications:
 - **Cosine Similarity:** Measures the cosine of the angle between two vectors, ideal for text and semantic similarity tasks.
 - **Euclidean Distance (\$L_2\$):** Measures the straight-line distance between two points in the vector space, often used for computer vision applications.
- **Metadata Keys:** Users can define up to 10 keys for storing associated metadata with each vector. This metadata can be used for filtering queries, allowing for hybrid search strategies (e.g., "find vectors similar to this one, but only for documents created after a certain date"). The total size of the metadata for a single vector is limited to approximately 40 KB.

Each individual index is capable of storing up to 50 million vectors. For datasets exceeding this limit, a sharding strategy across multiple indexes would be required.

3.1.3 Native Indexing and Search API

A key advantage of S3 Vectors is that it completely abstracts away the complexity of vector indexing. Users do not need to choose or tune complex Approximate Nearest Neighbor (ANN) algorithms. S3 Vectors automatically handles the indexing process internally, optimizing the data structures for the best price-performance balance as the dataset scales.

Interaction with the service is managed through a dedicated set of APIs for inserting, querying, updating, and deleting vectors. The query API supports k-Nearest Neighbor (k-NN) similarity searches, allowing users to submit a query vector and retrieve the 'k' most similar vectors from the index based on the configured distance metric. The service is designed to deliver sub-second query performance for these searches. However, it's important to note

that write operations are throttled, with a limit of approximately five requests per second per index, making the service better suited for read-heavy or nearline workloads rather than high-frequency, real-time data ingestion scenarios.

3.2 Seamless Integration with the AWS Ecosystem

One of the most compelling aspects of S3 Vectors is its deep, native integration with other key services in the AWS AI/ML ecosystem, creating a cohesive and powerful platform for building intelligent applications.

- **Amazon Bedrock:** S3 Vectors serves as a first-class knowledge base option for Amazon Bedrock, AWS's managed service for foundation models. This integration dramatically simplifies the development of RAG applications. Users can point their Bedrock Knowledge Base directly to an S3 Vector bucket, which then handles the entire retrieval part of the RAG workflow automatically, allowing foundation models to answer questions using private, proprietary data stored and indexed in S3.
- **Amazon SageMaker:** For data scientists and ML practitioners, S3 Vectors integrates with Amazon SageMaker Studio. This allows users to build, manage, and query vector knowledge bases directly within their familiar development environment, streamlining the MLOps lifecycle from model training and embedding generation to vector indexing and search.
- **Amazon OpenSearch Service:** S3 Vectors is designed to work in concert with Amazon OpenSearch Service to enable sophisticated, cost-effective tiered storage architectures. In this pattern, S3 Vectors can act as a massive, low-cost "cold" or "warm" tier for storing billions or trillions of vectors that are accessed less frequently. The most frequently accessed "hot" data can be moved to OpenSearch for high-throughput, millisecond-latency queries. This hybrid approach allows architects to precisely balance performance and cost across their entire vector dataset.

This serverless, fully managed, and deeply integrated approach positions Amazon S3 Vectors not merely as a feature, but as a foundational component of the modern, AI-ready data architecture on AWS.

4. The Value Proposition: Key Benefits and Advantages of S3 Vectors

The introduction of Amazon S3 Vectors provides a compelling value proposition that addresses the core challenges of traditional vector search architectures. The benefits extend beyond simple feature additions, offering fundamental improvements in architectural design, cost efficiency, operational simplicity, and development velocity. For solutions architects and engineers, these advantages translate into the ability to build more powerful, scalable, and economically viable AI applications.

4.1 Radical Architectural Simplification

The most significant benefit of S3 Vectors is the dramatic simplification of the AI application stack. By eliminating the need for a separate, dedicated vector database, S3 Vectors collapses a complex, multi-layered architecture into a streamlined, unified data layer.

Before S3 Vectors, a typical architecture involved:

1. An Amazon S3 bucket for raw object storage.
2. A complex ETL pipeline to monitor S3, generate embeddings, and synchronize data.
3. A separate vector database for storing and indexing the embeddings.
4. Application logic to query the vector database and then retrieve the corresponding object from S3 using a stored reference.

With S3 Vectors, the architecture becomes:

1. An Amazon S3 Vector Bucket to store both the source data (as objects) and the vector embeddings.
2. A simplified ETL process that generates embeddings and writes them directly to the vector index within the same S3 ecosystem.
3. Application logic that queries the S3 Vector index directly.

This consolidation reduces the number of moving parts, eliminates complex synchronization logic, and minimizes potential points of failure. As a result, teams can focus their valuable engineering resources on developing core application features and improving ML models rather than managing and integrating disparate infrastructure components.

4.2 Unprecedented Cost Reduction

S3 Vectors is engineered to deliver a transformative reduction in the total cost of ownership (TCO) for vector workloads. AWS claims it can reduce the cost of storing and querying vectors by up to 90% compared to traditional vector database solutions. This dramatic cost saving is achieved through several key architectural and pricing model decisions:

- **Leveraging S3 Storage Economics:** The service is built on S3's hyper-scale, cost-effective storage infrastructure. Storage pricing for vectors and their metadata is significantly lower than the RAM-based storage typically required by high-performance vector databases.
- **Pay-Per-Use Pricing Model:** Unlike traditional databases that often require provisioning expensive compute instances that incur costs even when idle, S3 Vectors employs a serverless, pay-per-use model. Customers are charged only for the storage they consume, the number of vectors they insert, and the amount of data processed during queries. This eliminates the cost of idle capacity and makes the service exceptionally cost-effective for workloads with variable or infrequent query patterns.
- **Separation of Storage and Compute:** The architecture's separation of storage and compute allows for independent scaling, which is far more efficient. You can store massive amounts of vector data cheaply without needing to provision correspondingly massive and expensive compute resources if your query volume is low to moderate.

For workloads with low to medium query volumes on medium-sized datasets, S3 Vectors can be 30 to 130 times cheaper than hosted vector database alternatives. This makes advanced AI capabilities financially accessible to a much broader range of projects and organizations.

4.3 Massive Scalability and Enterprise-Grade Durability

By being built directly on the foundation of Amazon S3, S3 Vectors inherits the legendary scalability and durability of its parent service.

- **Scalability:** S3 is designed for exabyte-scale data, and S3 Vectors leverages this capability to store billions of vectors without issue. The serverless nature of the service means that it scales automatically to handle growing datasets and query loads without requiring any manual intervention or capacity planning.
- **Durability:** S3 is designed for 11 nines (99.999999999%) of data durability, meaning that data stored in S3 Vectors is protected against loss with an extremely high degree of

reliability. This provides peace of mind and is critical for enterprise applications where data integrity is paramount.

- **Consistency:** The service provides strong consistency for data operations, ensuring that once a write is confirmed, it is immediately visible to all subsequent reads.

4.4 Accelerated AI Development and Innovation

The operational simplicity of S3 Vectors acts as a powerful accelerator for AI development. With no infrastructure to provision, configure, or maintain, development teams can begin building and experimenting with AI-powered applications almost immediately. This agility is a significant competitive advantage. The low barrier to entry and reduced costs encourage experimentation, allowing teams to explore new use cases for vector search without committing to a large upfront investment in infrastructure and specialized expertise. The seamless integration with services like Amazon Bedrock further shortens the path from concept to production for complex applications like RAG.

In summary, the value proposition of S3 Vectors is a powerful combination of architectural elegance, dramatic cost savings, massive scale, and development agility. It fundamentally changes the calculus for implementing vector search, transforming it from a complex, specialist task into a broadly accessible, mainstream cloud capability.

5. Primary Use Cases and Implementation Patterns

Amazon S3 Vectors is a versatile tool designed to power a wide array of AI/ML workloads, particularly those where scalability and cost-efficiency are paramount. Its unique characteristics make it an ideal choice for several key implementation patterns that are becoming central to modern enterprise AI strategies.

5.1 Retrieval-Augmented Generation (RAG)

RAG has emerged as a leading architectural pattern for enhancing the capabilities of Large Language Models (LLMs) by grounding them in factual, proprietary, or real-time data. S3 Vectors is exceptionally well-suited to serve as the vector store, or "knowledge base," in a RAG architecture.

Implementation Pattern:

1. **Data Ingestion and Embedding:** Enterprise documents are ingested, chunked into manageable segments, and converted into vector embeddings using a foundation model.
2. **Vector Storage:** These embeddings, along with metadata (e.g., source document ID), are stored in an S3 Vector index.
3. **Retrieval and Generation:** When a user poses a query, the query is first converted into an embedding. This query vector is used to perform a similarity search against the S3 Vector index to retrieve the most relevant document chunks. These retrieved chunks are then passed to an LLM as context, along with the original prompt, allowing the model to generate a factually grounded and accurate response.

Advantage: The tight integration with Amazon Bedrock Knowledge Bases makes this entire workflow nearly seamless. Organizations can build highly scalable and remarkably cost-effective RAG applications for use cases like intelligent chatbots, document Q&A systems, and automated research assistants, without managing a separate vector database.

5.2 Enterprise Semantic Search

S3 Vectors enables developers to build semantic search that understands user intent beyond simple keyword matching.

Implementation Pattern:

- **Document Search:** An entire corpus of enterprise documents can be embedded and indexed in S3 Vectors. A user can search for a concept, and the system will retrieve relevant documents even if they don't contain those exact keywords.
- **Code Search:** A repository of source code can be embedded to allow developers to search for code snippets based on their functionality rather than specific function names.
- **Multimedia Content Retrieval:** Vectors generated from images, videos, or audio files can be indexed, enabling searches based on content similarity.

Advantage: This provides a unified search experience across diverse unstructured data types, backed by S3's scalability and cost structure, making it feasible to index vast corporate data lakes.

5.3 AI Agent Long-Term Memory

For sophisticated AI agents and chatbots, maintaining context over long conversations is a significant challenge. S3 Vectors offers a cost-effective solution for creating a "long-term memory" for these agents.

Implementation Pattern:

1. **Conversation Logging:** As an AI agent interacts with a user, key turns of the conversation are vectorized.
2. **Memory Storage:** These vectors are stored in an S3 Vector index associated with the user's profile.
3. **Contextual Retrieval:** At the beginning of a new interaction, the agent can query its memory store to retrieve relevant past context, enabling more personalized and coherent conversations.

Advantage: Compared to storing conversation histories in a more expensive, memory-intensive database, S3 Vectors provides a highly scalable and economical solution for persisting long-term context, especially for applications with millions of users.

5.4 Multimedia Similarity Search and Analysis

By using powerful multimodal embedding models, S3 Vectors can power a variety of visual and audio search applications.

Implementation Pattern:

- **Medical Imaging:** A hospital can build a system that allows radiologists to find historical medical images that are visually similar to a patient's current scan, aiding in diagnosis.
- **Copyright Infringement Detection:** Content creators can index their visual or audio assets. When new content is uploaded, its embedding can be checked against the index to detect potential copyright violations.
- **Product Recommendation:** E-commerce platforms can use visual similarity to recommend products based on an uploaded photo.

Advantage: S3's ability to store large media files and their vector embeddings together simplifies architecture for media-heavy applications.

5.5 Cost-Optimized Tiered Data Architectures

For applications with massive vector datasets and diverse access patterns, S3 Vectors is a perfect component for a tiered, or hybrid, data architecture.

Implementation Pattern:

1. **Cold/Warm Tier:** The vast majority of vectors, especially those that are accessed infrequently, are stored in S3 Vectors. This provides extremely low-cost storage with acceptable sub-second query latency for occasional access.
2. **Hot Tier:** The most frequently accessed or latency-sensitive vectors are replicated to a high-performance system like Amazon OpenSearch Service.
3. **Data Lifecycle Management:** A data lifecycle policy can be implemented to automatically move data between the hot and cold tiers based on access patterns, ensuring an optimal balance between performance and cost.

Advantage: This pattern allows organizations to manage petabyte-scale vector datasets in a financially sustainable way. It prevents the need to store the entire dataset in expensive, high-performance memory.

6. Limitations

Before examining the internal performance characteristics and design trade-offs of Amazon S3 Vectors, it is critical to establish a clear understanding of the platform's inherent limitations. These boundaries define where the service excels, where it falls short, and what technical and operational assumptions architects must make before building on top of it. As a relatively young service still in preview, Amazon S3 Vectors balances innovation and constraint—pioneering new cloud-scale vector search capabilities while placing reasonable limits on flexibility, configurability, and performance.

6.1 Preview Release and Service Maturity

Amazon S3 Vectors is currently in **preview release**, meaning that it remains an experimental service within Amazon's portfolio. As with most preview offerings, users should expect evolving functionality, periodic API adjustments, and regional availability that does not yet encompass the full AWS global footprint.

While the service is stable enough for pilot workloads, proof-of-concept integrations, and internal testing, mission-critical systems requiring formal SLAs or multi-region guarantees should treat S3 Vectors as pre-production. Architects are advised not to tightly couple key business processes to current API behaviors or assumed performance benchmarks. Instead, designs should encapsulate abstraction layers or wrappers in their data pipelines, ensuring that future adjustments or migration to general availability versions require minimal code refactoring.

6.2 Account, Region, and Availability Limitations

During the preview period, **regional availability** is limited to selected zones such as **US East (N. Virginia)** and **EU (Frankfurt)**. For globally distributed organizations, this poses challenges in three main areas:

- **Regulatory compliance:** Jurisdictions with strict data locality laws (e.g., EU, China, Singapore) may not allow embeddings to reside outside regional boundaries.
- **Latency optimization:** Applications that must serve users in Asia-Pacific may experience higher network latency when all vector operations route through a U.S. or European region.

- **Disaster recovery and multi-region failover:** Preview deployments do not yet include automated cross-region replication, reducing options for active-active redundancy.

At the **account level**, S3 Vectors supports up to **10,000 vector buckets** and **10,000 vector indexes per bucket**. These generous thresholds enable massive deployments encompassing billions of vectors, but they still demand structured namespace management. For high-complexity enterprises, governance mechanisms such as tagging policies or AWS Organizations service control policies (SCPs) can help prevent accidental over-allocation or uneven resource distribution.

6.3 Index Capacity and Vector Configuration

At the core of every vector search workload lies the individual index, and here S3 Vectors introduces several key boundaries:

- Each index supports a maximum of **50 million vectors**.
- Vector **dimensions must be between 1 and 4096**.
- All vectors are of the **float32** data type.
- The **distance metric** and **dimension size** are immutable after index creation.

These restrictions define an upper bound for both storage and flexibility. While 50 million vectors can address many enterprise and AI workloads, massive organizations hosting multi-billion-embedding datasets—common in global e-commerce, recommendation, or digital twin use cases—must design sharding strategies to distribute vectors across multiple indexes. This introduces the need for additional orchestration logic at the application layer to route queries, merge similarity results, or deduplicate overlaps.

The **data type constraint** to float32 ensures consistent mathematical precision across embeddings generated by model frameworks such as PyTorch or TensorFlow. However, systems optimized for **quantized data formats** (e.g., int8, binary) cannot exploit the compression and speed benefits such encoding offers, making S3 Vectors less efficient in ultra-large deployments where hardware-level memory efficiency is paramount.

The **immutability** of index parameters—dimension size, distance metric, and metadata key schema—reinforces predictability and stability at the cost of flexibility. Once created, an index cannot be modified, meaning any change in embedding model size or metadata structure requires complete reingestion.

6.4 Metadata and Filtering Restrictions

Metadata enrichment allows semantic filtering alongside vector similarity scoring. However, in S3 Vectors the filtering model is intentionally simplified to support basic use cases without introducing complex query overheads.

Each index supports:

- Up to **10 metadata keys**.
- Key-value pairs limited in size per vector.
- Relatively basic filter logic (simple equality or comparative conditions).

While sufficient for use cases like “region = EU” or “category = finance,” it cannot express more involved Boolean logic, nested queries, or compound ranking based on multiple metadata dimensions. Applications requiring high composability—such as content recommendation systems combining metadata filters with user segmentation criteria—will need an auxiliary metadata store like Amazon DynamoDB or OpenSearch, with S3 Vectors reserved solely for raw vector similarity computation.

6.5 Write Throughput and Update Model Limitations

Perhaps the most significant operational limit is **low write throughput**. S3 Vectors currently supports roughly **five write requests per second per index**. This throughput caps the rate of ingestion, making it unsuitable for workloads that continuously stream embeddings or require real-time updates. S3 Vectors enables developers to build semantic search that understands user intent beyond simple keyword matching.

Optimal scenarios include **write-once, read-many (WORM)** or **batch ingestion** patterns, such as:

- Uploading document embeddings during nightly or weekly re-indexing jobs.
- Populating semantic stores for RAG applications where data refresh occurs periodically.
- Bulk loading datasets during onboarding phases followed by long-term query-focused usage.

In contrast, **dynamic pipelines**—for instance, social media feeds, fraud detection models, or recommendation engines updating embeddings per user interaction—would exceed throughput ceilings quickly. These must integrate alternative ingestion patterns such as buffering writes, asynchronous batches, or hybrid architectures combining fast in-memory

stores (e.g., Redis Vector Search) for real-time updates and S3 Vectors for long-term storage.

6.6 Latency and Query Performance Boundaries

Query operations achieve **sub-second latency**, typically falling within **100–800 ms** per request. While these figures are excellent for most semantic search or AI augmentation workloads, they fall short of **ultra-low-latency thresholds** demanded by live inference systems where responses must occur within a few milliseconds.

Thus, S3 Vectors aligns best with use cases such as:

- Retrieval-Augmented Generation (RAG) pipelines supporting LLM queries.
- Internal enterprise search across large document or knowledge bases.
- Semantic clustering and content similarity analysis.

But it is not ideal for:

- Real-time personalization engines.
- Fraud scoring or on-the-fly risk assessment.
- Edge-device decisioning where predictability under 50 ms is necessary.

In practical deployment, applications can cache vector results locally or use smaller index shards for active datasets while offloading long-term archives to S3 Vectors.

6.7 API and Data Size Constraints

Each API call is subject to maximum request payload and per-index concurrency limits. These quotas safeguard overall system stability across AWS's shared multitenant infrastructure. Bulk upload operations must chunk data into smaller packages and implement retry mechanisms with exponential backoff to comply with API throughput ceilings.

AWS SDKs, particularly boto3 and AWS CLI extensions, simplify these workflows through managed pagination and bulk-batch abstractions. Proper attention to API limits ensures that ingestion remains efficient and fault-tolerant, even when jobs span billions of vectors.

6.8 Summary of Limitations

Category	Key Limitation	Design Implication
Preview Status	Inconsistent availability and API evolution	Adopt abstraction layers to mitigate future changes
Regional Availability	Limited to selected regions	Restricts multi-region, compliance-sensitive workloads
Index Capacity	Max 50M vectors per index	Requires manual sharding for larger datasets
Vector Data Type	Float32 only	Larger storage footprint vs quantized vectors
Metadata Support	Max 10 keys, simple filters	Limited expressiveness; consider external metadata store
Write Throughput	~5 writes/sec per index	Only suitable for batch or write-once usage patterns
Latency	100–800 ms query range	Not suitable for real-time inference
Immutability	Index parameters fixed post-creation	Careful upfront schema planning required
API Limits	Constrained payload size	Implement batching and retry logic in ingestion code

7. Technical Considerations and Performance Profile

With the service limitations clarified, it becomes easier to understand the underlying rationale and architectural considerations that govern Amazon S3 Vectors' technical design. The following sections explore the operational envelope—how the service behaves under expected workloads, where trade-offs are made for scalability and cost, and what patterns emerge when building production architectures around it.

7.1 Architectural Philosophy

S3 Vectors extends the durability and scalability of Amazon S3 into vectorized data space. By embedding vector storage and similarity-search performance directly into S3's core infrastructure, AWS achieves a paradigm shift: massive-scale vector indexing without the operational complexity of managing dedicated clusters or databases.

This approach trades the speed of high-performance, in-memory vector engines for linear, globally scalable storage and retrieval performance across petabyte-scale datasets. It prioritizes **operational simplicity, global reach, and predictable cost efficiency** over extreme latency optimization.

7.2 Query Latency and Throughput Performance

Under typical conditions, **latency averages between 100–800 ms**, influenced by data distribution, vector dimensionality, and the use of metadata filters. Compared to purpose-built systems such as Milvus or Pinecone—often achieving single-digit millisecond responses—S3 Vectors trades raw performance for elastic scale and cost regularity.

For workloads like:

- LLM retrieval in RAG frameworks (LangChain, Bedrock, SageMaker JumpStart).
- Semantic similarity across document or image collections.
- Search augmentation within customer support or enterprise knowledge bases.

this trade-off remains acceptable. However, applications demanding continuous ingestion or live computation must account for latency budgets and structure asynchronous flows where vector operations run independently from synchronous user interactions.

7.3 Storage Design and Index Management

Indexes built within S3 Vectors are immutable configurations defining vector dimensionality, distance metric, and metadata schema. Given these characteristics, each index operates as a stable container suitable for **long-term semantic repositories**—in essence, an AI-native data lake structure.

Architects should:

- Separate **active datasets** (frequently queried indexes) from **archival datasets** to control query overhead.
- Employ **batch processing pipelines** (AWS Glue, Step Functions) for controlled write scheduling within throughput limits.
- Implement **naming schemes** and **metadata version control** to maintain traceability of embeddings across reindexed generations.

Because S3 Vectors inherits S3's resiliency guarantees (11 nines durability), it naturally extends into compliance-driven sectors needing resilient AI infrastructure—without cluster scaling overhead.

7.4 Data Type and Computation Efficiency

All embeddings are stored as **float32 arrays**, offering precision consistent with mainstream deep learning models. This uniformity ensures interoperability across multiple frameworks. Nevertheless, float32 storage implies higher byte consumption—approximately 4 bytes per dimension.

This leads to predictable but potentially costly storage footprints, e.g.:

$50,000,000 \times 1024 \times 4 \text{ bytes} = 204.8 \text{ GB}$
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per 50-million-vector index (excluding metadata and operational overhead). While this is manageable under modern S3 economics, users should monitor cost when scaling across thousands of indexes.

7.5 Configuration Immutability Considerations

The immutability of index parameters anchors consistency and prevents unintentional data drift but limits iterative model development. Teams exploring different embedding models or similarity metrics must adopt **clone-and-recreate strategies**, automating index provisioning and reingestion.

AWS-native automation stacks—CloudFormation, Step Functions, or Terraform modules—can recreate multiple index versions synchronized with model versioning pipelines. This enables continuous improvement of vector quality while maintaining reproducible infrastructure.

7.6 Metadata Filtering and Query Semantics

Metadata in S3 Vectors acts as a lightweight prefilter mechanism. Engineers designing search pipelines should treat metadata as *index-level partitions* that narrow down the candidate space prior to similarity scoring. Optimizing metadata design—e.g., using coarse categories rather than micro-level tags—reduces latency while staying within the 10-key limitation.

For complex search semantics requiring logical composition (e.g., “region = APAC AND topic IN [finance, insurance]”), organizations can:

- Store metadata separately in Amazon DynamoDB or OpenSearch.
- Retrieve candidate IDs based on metadata filters.
- Execute vector similarity search only on relevant subsets.

This hybrid approach ensures both efficient filtering and flexible business rule expressiveness.

7.7 Scalability Planning

S3 Vectors' **50M vectors per index** limit drives the need for **application-level sharding** once datasets exceed that size. Best practices include:

- Logical grouping of vectors by domain, language, or source system.
- Query fan-out and aggregation logic using AWS Lambda or API Gateway orchestration.
- Weighted score merging to unify results across shards.

This architecture pattern resembles large search index federations, where multiple indexes collectively simulate one massive store.

7.8 Balancing Cost, Latency, and Usability

The true value of S3 Vectors lies in its **cost/performance equilibrium**. Compared to high-performance vector databases that require persistent clusters (with GPU or memory-based architectures), S3 Vectors' pay-per-use model significantly lowers total ownership costs, particularly for workloads where queries are frequent but not continuous.

Optimization recommendations:

- Schedule batch ingestion during off-peak hours to minimize cost impact.
- Use caching layers (e.g., Elasticache or CloudFront edge caches) to offset moderate query latencies.
- Periodically evaluate vector dimensionality and pruning strategies to balance precision and storage footprint.

7.9 Regional and Regulatory Considerations

Until full global rollout, organizations with compliance obligations should maintain geofenced architectures. A hybrid approach—partitioning embeddings regionally and using **AWS PrivateLink** for internal routing—can achieve data locality while minimizing cross-region latency.

Summary: Strategic Perspective

Amazon S3 Vectors situates itself between traditional vector databases and scalable cloud object storage. It offers unprecedented integration simplicity, immense scale, and favorable economics, while imposing boundaries around latency, write throughput, and configurability.

Technical Dimension	Key Strength	Core Trade-Off
Scalability	Virtually unlimited global reach via S3 fabric	Manual sharding beyond 50M vectors per index
Operational Model	Fully managed, no servers or clusters	Limited configurability, immutable indexes
Cost Efficiency	Pay-as-you-go using S3 pricing	Potential storage inefficiency for float32 embeddings
Performance	Sub-second latency for semantic workloads	Insufficient for real-time inference
Integration	Works natively with Bedrock, Lambda, SageMaker	Limited in advanced metadata filtering or schema evolution

For enterprises modernizing their search and AI ecosystems, S3 Vectors introduces a **durable, scalable, and low-complexity foundation**. It is particularly effective for AI knowledge stores, RAG retrieval layers, and machine learning pipelines emphasizing cost predictability and cross-organization scalability.

8. Architectural Best Practices and Integration Patterns

Designing production-grade vector and AI search solutions with Amazon S3 Vectors requires thoughtful orchestration across multiple AWS building blocks. Although the service simplifies vector storage management, enterprise-grade deployment still demands careful coordination of data pipelines, indexing workflows, and inference layers.

This section outlines proven patterns and practical blueprints for integrating S3 Vectors with complementary AWS services including **Bedrock**, **Lambda**, **SageMaker**, and **Glue**, illustrating how each contributes to an efficient, scalable retrieval-augmented architecture.

8.1 Architecture Overview

At the highest level, an S3 Vectors deployment can be organized into four interacting layers:

1. **Ingestion and Embedding Layer**
Responsible for extracting raw data, generating embeddings via machine-learning models, and writing them to vector indexes or batches.
2. **Vector Storage and Index Layer**
Managed by S3 Vectors itself—storing float32 embeddings along with limited metadata for similarity search.
3. **Retrieval and Reasoning Layer**
Uses AWS Bedrock or SageMaker endpoints to execute queries, perform semantic search, and feed results to downstream systems or applications.
4. **Application and Integration Layer**
Includes calling services such as API Gateway, AWS Lambda, or containerized microservices that wrap query logic and deliver responses to users or applications.

Each layer can scale independently using AWS's serverless or managed toolset, ensuring cost elasticity and minimal administrative overhead.

8.2 Best Practices for Data Preparation and Embedding

Creating high-quality embeddings is fundamental to effective similarity search. Follow these practices to optimize embedding quality and consistency:

- **Choose consistent models:**
Standardize on a single embedding model family (e.g., Amazon Titan Embeddings via Bedrock, OpenAI text-embedding-3-large, or custom SageMaker JumpStart models) to ensure dimension uniformity.
- **Normalize and deduplicate input data:**
Remove redundant records and standardize text cleaning, tokenization, and lower-casing before embedding generation.
- **Batch embedding jobs:**
Use AWS Glue or Step Functions to run embedding generation at scale. Write completed embeddings to temporary S3 objects before ingestion to S3 Vectors to stay within API write-throughput limits.
- **Track model metadata:**
Store embedding model ID, version, and timestamp in accompanying metadata fields for reproducibility and auditability.

8.3 AWS Lambda as the Orchestration Layer

AWS Lambda provides the elastic execution backbone that binds indexing and query operations together. Recommended configurations include:

- **Batch Ingestion Pipeline**
 - Trigger Lambda from S3 upload events to process new documents.
 - Generate embeddings using Bedrock or SageMaker, then batch-write vectors via S3 Vectors API.
 - Handle retries with exponential backoff respecting the five-writes-per-second limit.

- **Vector Query Wrapper**

- Expose semantic search via an API Gateway endpoint linked to Lambda.
- Lambda accepts natural-language queries, calls the embedding endpoint to vectorize input, and invokes the S3 Vectors Search API.
- Results are filtered and returned to users or further composed with LLM completions in Bedrock.

This modular Lambda-based pattern isolates compute execution from data storage, ensures stateless scalability, and promotes clean separation of responsibilities.

8.4 Leveraging Amazon Bedrock for RAG Workloads

Amazon Bedrock natively integrates with multiple foundation models and, when combined with S3 Vectors, forms a robust RAG (Retrieval-Augmented Generation) architecture.

Workflow pattern:

1. A user query arrives through API Gateway or an application interface.
2. A Lambda function converts the query into an embedding using Bedrock Titan Embeddings.
3. Lambda issues a similarity search to S3 Vectors and retrieves the top-K most relevant documents.
4. The same Lambda or another Bedrock agent composes the retrieved context as an augmented prompt and sends it to the chosen LLM (e.g., Claude 3 on Bedrock).
5. The LLM synthesizes a contextual answer, referencing the retrieved material.

Optimization tips:

- Cache frequent embedding requests in DynamoDB Accelerator (DAX) to reduce repeated vectorization latency.
- Use Bedrock Agents to manage RAG flow declaratively rather than through custom code.
- Pre-compute embeddings offline for static corpora, updating only changed content during scheduled jobs.

8.5 Training and Custom Refinement with SageMaker

For enterprises developing proprietary embedding models or domain-specific similarity logic:

- **Model Training:**
Train sentence-embedding or multimodal models on SageMaker Training Jobs using domain data. Export trained weights to S3 for reproducibility.
- **Batch Embedding Generation:**
Run large-scale inference with SageMaker Batch Transform directly writing embeddings into staging S3 buckets.
Integrate a Step Function that uploads these batches to S3 Vectors respecting throughput quotas.
- **Evaluation and Monitoring:**
Implement MLOps workflows via SageMaker Pipelines to measure retrieval recall, precision, and latency over time, automatically retraining if drift exceeds thresholds.

This allows the embedding logic to evolve while maintaining storage continuity in S3 Vectors indexes.

8.6 Metadata Management and Auxiliary Indexing

Given S3 Vectors' 10-key metadata constraint, extended metadata should reside in auxiliary stores:

- **Amazon DynamoDB** for lightweight attribute filtering and timestamp tracking.
- **Amazon OpenSearch Service** for full-text search and result boosting prior to vector recall.
- **Amazon Athena** on top of S3 data for analytical queries correlating embeddings with structured attributes.

Architecturally, each vector record can carry a unique ID that links its extended metadata in these companion systems. Combined queries merge structured filters and vector similarity ranks at the application layer.

8.7 Sharding and Query Federation Patterns

To overcome the **50 million-vector-per-index** ceiling, implement **application-level sharding**.

Implementation approaches:

- Divide data by source, geography, or embedding model version.
- Store shard metadata (index ARNs, categories) in DynamoDB.
- Build query-fan-out logic using Lambda, Step Functions, or Amazon Kendra Connectors to simultaneously query multiple indexes and aggregate results.
- Normalize similarity scores across shards to maintain ranking consistency. This pattern provides elastic horizontal scaling without centralized re-indexing costs.

8.8 Security and Compliance Practices

Security inherits all S3 strengths while adding vector-specific controls.

- **IAM-based least-privilege design:** Restrict access to `s3vectors:Search`, `s3vectors:WriteIndex`, and `s3vectors:ReadIndex` actions only for required roles.
- **Encryption:** Leverage server-side encryption with AWS KMS; specify customer-managed keys when embedding sensitive data.
- **Network control:** Access S3 Vectors over VPC Endpoints using AWS PrivateLink to avoid public internet exposure.
- **Data locality:** Maintain region-specific buckets for jurisdictions subject to GDPR-style regulations.

Embedding security posture and compliance considerations from S3's architecture ensures continuity for enterprise governance frameworks.

8.9 Monitoring, Logging, and Cost Optimization

Robust observability helps manage large deployments while controlling expenditure.

- **Logging:** Enable CloudTrail for API calls and CloudWatch Logs for indexing and query latency metrics.
- **Metrics:** Track average query latency, search volume, index utilization, and error rates to detect anomalies early.
- **Cost Insights:** Combine AWS Cost Explorer and CloudWatch dashboards to identify seldom-queried indexes eligible for lifecycle archiving or consolidation.
- **Lifecycle Policies:** Integrate S3 Object Lifecycle rules to automatically transition old embeddings to lower-cost tiers like Glacier Instant Retrieval if inactive.

8.10 Example End-to-End Reference Architecture

1. Data Ingestion and Embedding Generation

News articles arrive in S3 raw bucket → Glue Job cleanses → SageMaker Batch Transform generates embeddings → Batch results staged in intermediate S3 folder.

2. Index Creation and Population

A Step Function workflow creates an S3 Vector index with dimensional metadata → Lambda batches write requests (≤ 5 per second) → Progress logs stored in CloudWatch.

3. Query Serving Path

User submits query through API Gateway → Lambda obtains embedding from Bedrock Titan → Lambda performs **SearchIndex** on S3 Vectors → Top-K contexts returned → Bedrock LLM composes final response → User receives enriched answer.

This reference flow delivers near-real-time semantic search with minimal administrative maintenance, leveraging native AWS services end-to-end.

8.11 Future-Ready Design Considerations

As S3 Vectors matures toward general availability, new features such as additional regions, binary vector support, or improved write throughput may be introduced. To future-proof your architecture:

- Abstract the S3 Vectors API behind internal SDKs or interface classes to accommodate evolving schemas.
- Use environment configuration files (e.g., Parameter Store, Secrets Manager) for index ARNs and embedding model identifiers.
- Decouple data pipelines so embedding generation and vector storage can evolve independently.
- Regularly validate latency, recall, and cost metrics against expected SLAs.

Building around flexible patterns today ensures seamless upgrades tomorrow without disruptive rewrites.

8.12 Synthesis: Designing for Scale, Simplicity, and Continuity

Amazon S3 Vectors is not positioned as a competitor to high-throughput, sub-millisecond vector databases but rather as a **cloud-native vector lake**—an infrastructure layer merging durability, scalability, and low operational overhead. Integrating it with Bedrock’s LLM orchestration, SageMaker’s model lifecycle, and Lambda’s event automation empowers organizations to:

- Centralize vast semantic datasets.
- Simplify AI retrieval pipelines.
- Lower infrastructure costs through serverless resource allocation.
- Maintain governance and compliance without sacrificing agility.

By combining these AWS components coherently, enterprises can deploy retrieval-augmented systems that are both operationally resilient and economically optimized, paving a path for scalable AI applications across departments, geographies, and business lines.

9. Conclusion: Redefining AI-Ready Storage for the Enterprise

Amazon S3 Vectors represents more than an incremental feature update; it marks a significant and strategic evolution of cloud storage, fundamentally repositioning Amazon S3 as a foundational, AI-native layer for the modern data stack. By embedding powerful vector search capabilities directly into the fabric of the world's most ubiquitous object storage service, AWS has effectively democratized access to a technology that was once the domain of specialized, complex, and costly systems. This move radically simplifies the architecture for a new generation of intelligent applications, making scalable and sophisticated AI more accessible, affordable, and easier to implement for organizations of all sizes.

For solutions architects and AI engineers, S3 Vectors provides a compelling new building block. It offers a streamlined path to developing robust, enterprise-grade applications for Retrieval-Augmented Generation, semantic search, and large-scale data analysis, backed by the unparalleled durability and scalability of S3. The dramatic reduction in total cost of ownership, driven by a serverless architecture and pay-per-use pricing model, changes the economic equation for storing and querying massive vector datasets, unlocking new opportunities for innovation that were previously cost-prohibitive.

However, it is crucial to recognize that S3 Vectors is not a panacea. Its design makes explicit trade-offs, prioritizing cost-efficiency and scale over the ultra-low latency and high write throughput characteristic of specialized, in-memory vector databases. It is a purpose-built solution, masterfully engineered for the nearline, read-heavy workloads that dominate many RAG and analytics use cases. For applications demanding real-time responsiveness in the millisecond range, a hybrid architecture combining S3 Vectors as a cost-effective capacity tier with a high-performance system like Amazon OpenSearch Service represents a powerful and balanced design pattern.

As artificial intelligence and machine learning continue their inexorable march to reshape every industry, the ability to efficiently harness the value within vast reserves of unstructured data will be a key differentiator. Amazon S3 Vectors provides the robust, scalable, and cost-effective foundation necessary to build this future. It empowers technical professionals to move faster, experiment more freely, and build smarter, more capable applications on the AWS cloud, cementing S3's role not just as a data lake, but as an intelligent, AI-ready data engine for the years to come.